

Magnetic Resonance-guided Focused Ultrasound (MRgFUS) for Brain Tumors and Brain Cancers

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Abstract

Magnetic resonance-guided focused ultrasound (MRgFUS) combines MRI guidance with focused ultrasound (FUS) to non-invasively treat brain tumors. This review examines its mechanisms, applications, and advantages over surgery and radiation. MRgFUS avoids craniotomy-related risks and ionizing radiation, enabling precise tumor targeting and repeatable treatments. Two primary mechanisms drive its therapeutic effects: thermal ablation via high-intensity FUS (HIFU) induces tumor necrosis, while mechanical cavitation via low-intensity FUS (LIFU) transiently opens the blood-brain barrier (BBB) for enhanced drug delivery. Current clinical applications include glioblastoma multiforme (GBM), astrocytomas, and brain metastases from other cancers, with trials demonstrating reduced neurological damage and improved chemotherapy uptake.

Despite its promise, challenges persist: skull-induced heating and ultrasound refraction limit access to peripheral tumors, treatment parameters require optimization, and long-term safety data remain sparse. Future advancements aim to expand MRgFUS utility through lower-frequency transducers for broader lesion coverage, combination therapies (chemotherapy, immunotherapy, nanoparticles), and exploitation of non-thermal effects like immune activation. More extensive multi-center trials are critical to validate efficacy across diverse tumor types and patient populations.

Keywords: MRgFUS, brain tumors, HIFU, LIFU, thermal ablation, cavitation, microbubbles

Introduction

In recent years, the escalating global incidence of brain tumors presents a pressing challenge to healthcare systems worldwide.¹⁻³ Malignant tumors of the central nervous system (CNS) are particularly devastating, not only as a primary cause of cancer-related mortality in children but also as a significant contributor to morbidity and mortality across all age groups.^[1] Despite years of scientific studies and clinical advancements, the prognosis for many brain cancers, most especially those aggressive forms like glioblastoma multiforme (GBM), remains unfavourable.¹⁻⁵ Current treatment strategies often combine surgery, radiotherapy, and chemotherapy, but these approaches have significant drawbacks.¹ Surgery poses risks such as infection, radiotherapy can harm nearby healthy tissues, and chemotherapy frequently causes severe side effects.^{[2][5][6]} Another critical obstacle in brain tumor treatments is the blood-brain barrier (BBB), a highly selective barrier restricting the passage of most chemotherapeutic agents, thereby hindering treatment efficacy and contributing to the resistance of many brain malignancies.¹⁻² The challenges associated with current brain tumor therapies highlight the need for innovative, less invasive, and more targeted treatment strategies to improve patient outcomes.^{4,6-7}

Magnetic resonance-guided focused ultrasound (MRgFUS) has emerged as a promising non-invasive and precise alternative approach for the treatment of brain tumors. MRgFUS integrates the therapeutic capabilities of focused ultrasound (FUS) with the real-time imaging guidance and temperature monitoring features of magnetic resonance imaging (MRI). With the precise delivery of focused acoustic energy through the intact skull to a targeted intracranial region, MRgFUS offers the potential to treat brain tumors with minimal off-target effects and without the need for surgical incisions or ionizing radiation.⁸⁻¹⁰ The non-invasive nature of MRgFUS translates to potential benefits such as reduced operative complications, shorter hospital stays, and repeatable treatments.¹¹

Background & History

The therapeutic potential of FUS in the brain was first demonstrated in 1942 by Lynn and colleagues.^{9,12} Their preclinical work utilized what they described as a “focused supersonic beam” to target brain tissue in animal models.¹² This study revealed that high-intensity focused ultrasound (HIFU) could produce precise localized tissue destruction.¹² Building on these early findings, Lynn and Putnam (1944) conducted detailed histological analyses of the FUS-induced cerebral lesions, confirming its ability to generate site-specific damage in neural tissue.¹³

In the 1950s, the Fry brothers were among the first to show that FUS could be used for precise neural ablation with minimal collateral damage to surrounding tissues.⁹ In a study published in 1958, Fry et al. demonstrated reversible alterations in the central nervous system, which laid the foundational groundwork for the use of FUS in functional neurosurgery.¹⁴ Their contributions helped shift the perception of FUS from a purely experimental tool to a promising modality for targeted brain interventions.

Despite these early breakthroughs, the clinical translation of FUS for neurosurgical applications progressed slowly over the following decades.^{9,11} One of the major technical challenges was the human skull, which significantly impedes the transmission of ultrasound waves.^{1,9,11} The skull's heterogeneous composition and its variable acoustic and absorptive properties created substantial barriers to effective sonication.^{1,9,11,15} Early efforts to deliver FUS to intracranial targets often necessitated invasive procedures, such as craniotomy (i.e., opening of the skull).^{1,9,11,16} Fry and Barger (1978) conducted important research into the acoustic characteristics of the human skull, identifying its role as a major obstacle in the development of noninvasive FUS techniques.^{11,17}

The integration of MRI into FUS systems marked a turning point in overcoming these limitations. MRI provided real-time visualization of anatomical targets, precise temperature

mapping via MR thermometry, as well as thermal feedback to guide energy delivery.^{1,9,16,18–19}

This innovation enabled accurate and non-invasive transcranial sonication, eliminating the need for surgical exposure of the brain.^{1,16} Hynynen and colleagues (1996) were among the first to explore MRI-guided focused ultrasound (MRgFUS) for clinical applications, proposing a method of noninvasive brain surgery monitored by MR imaging.²⁰

Subsequent research continued to refine the technology. Hynynen et al. (2004) introduced a 500-element ultrasound phased array system capable of targeting brain tissue noninvasively in a rabbit model using *ex vivo* human skulls.^{9,11,18} Aubry et al. (2003) demonstrated that computed tomography (CT)-based skull mapping could be used to compensate for skull-induced distortions, allowing for adaptive focusing during MRgFUS procedures.¹¹ This approach is now standard practice, with CT and MR images commonly fused during treatment planning.^{8,11,21}

The convergence of MRI guidance, phased-array transducers, and skull aberration correction enabled the clinical adoption of MRgFUS in the early 2000s.²² Initially applied to soft tissue tumors, MRgFUS quickly gained attention for its potential in functional neurosurgery due to its noninvasiveness and precision.^{1,8,11} One of the major institutions at the forefront of clinical MRgFUS research and application has been Sunnybrook Health Sciences Centre in Canada.^{23–24} At the same time, InSightec emerged as a leading developer of MRgFUS systems, notably the Exablate Neuro platform designed for transcranial applications.^{1,16,18}

With regard to regulatory milestones, the United States Food and Drug Administration (FDA) approved MRgFUS for the treatment of essential tremor in 2016.²⁵ Such a milestone represented a significant leap forward, facilitating wider clinical adoption and spurring a new wave of research. As of today, it is estimated that up to 2,000 patients in the United States alone have undergone MRgFUS procedures.⁸

Mechanisms of MRgFUS

MRi guidance

MRI supports accurate targeting, real-time thermal monitoring, and post-treatment evaluation in MRgFUS systems.^{7,9,18–19,26} High-resolution MRI (e.g., fMRI, perfusion imaging, and tractograph) allows for precise imaging of target tissues and surrounding critical structures.⁹ This level of precision is crucial for treatment planning to protect the major cortical regions and white matter tracts.⁹

During treatment, real-time MR thermometry—primarily through proton resonance frequency (PRF) shift—provides thermal feedback by mapping temperature changes within tissues.^{1,7,9,18–19,21–22} This ensures the target reaches therapeutic thresholds (i.e., above 56°C for ablation) while minimizing collateral damage.^{5,9,18,26}

Post-treatment MRI, particularly with contrast enhancement, verifies treatment effectiveness through pinpointing non-perfused ablated tissue and identifying potential complications such as hemorrhage or edema.^{7,9–10,16,18,22} Follow-up imaging, including diffusion and perfusion MRI, aids in long-term monitoring and assessment of therapeutic outcomes.^{7,9}

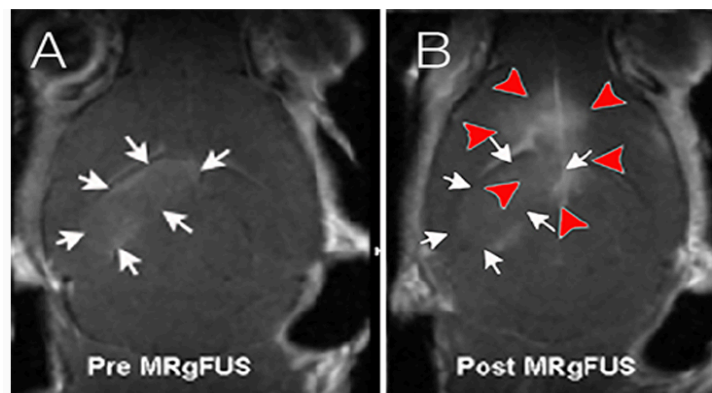


Figure 1. (A–B) Post-gadolinium T1-weighted MR images of a mouse brain showing a right-hemisphere tumor (arrows). After MRgFUS (B), contrast enhancement increases (arrowheads) due to BBB opening in both tumor and adjacent tissue.⁶

Focused ultrasound (FUS)

As the therapeutic arm of the MRgFUS system, FUS is capable of inducing both thermal and non-thermal biological effects in the targeted tissue. These effects are achieved by precisely concentrating ultrasound energy at a specific focal point deep within the tissue.

Thermal effects

The primary thermal mechanism of MRgFUS is the induction of coagulative necrosis through HIFU energy. When high-intensity waves converge at a target, their energy converts to heat, raising tissue temperatures above 56°C, which causes irreversible cell damage and targeted ablation without surgery.^{7,9,11,18–19,22,26}

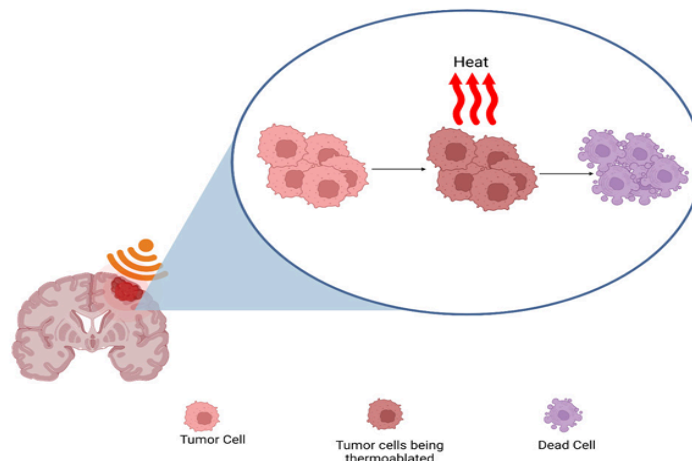


Figure 2. Schematic illustrating how continuous HIFU waves generate heat to induce protein denaturation and cell death.¹

Ultrasound parameters—intensity, frequency, and duration—are precisely controlled to optimize thermal effects. Intensities between 500–20,000 W/cm² determine heating rates, while frequencies of 0.5–1.1 MHz are typically used for transcranial applications to balance skull penetration and focal precision.^{1,7–9} Hemispherical phased arrays with hundreds of transducers (e.g., 1024 elements) allow for electronic steering and accurate targeting.¹⁸ Treatment settings are adjusted in real-time using MR thermometry to ensure safe and effective energy delivery.⁸

Non-thermal effects

The primary nonthermal mechanism of MRgFUS is cavitation, particularly inertial cavitation, which plays a major role in producing biological effects without significant tissue heating.^{9,18,26} Cavitation is induced by intravenously administered microbubbles (MBs) that circulate through brain capillaries.⁹

At low acoustic power, microbubbles (MBs) undergo stable cavitation, oscillating in size and generating shear forces that disrupt endothelial tight junctions, leading to temporary blood-brain barrier (BBB) opening.^{6,9} LIFU, which induces stable cavitation for BBB opening, typically uses frequencies ranging from 250 kHz to 650 kHz for transcranial applications in humans.^{1,7} One study also mentions the use of a lower frequency of 220 kHz for transient BBB opening.¹ At higher pressures, inertial cavitation occurs, where bubbles collapse violently, releasing shock waves that can damage tissue.^{7,9} While higher frequencies like 1.5 MHz have been used in preclinical studies, the lower frequency range is generally preferred for penetrating the human skull.^{1,7,21}

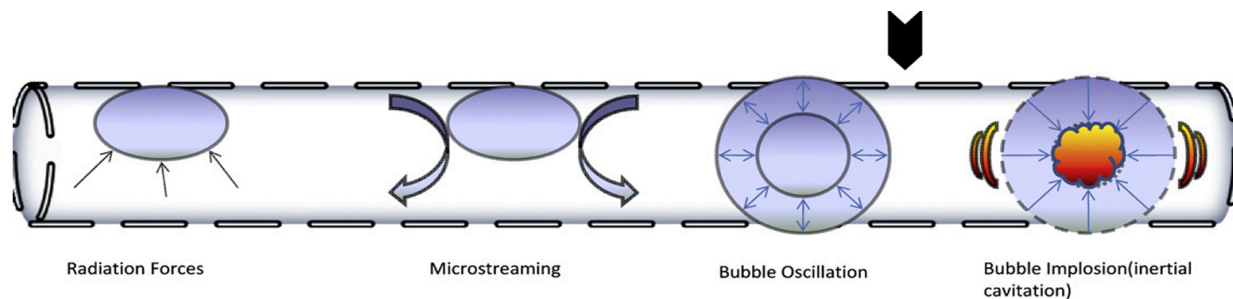


Figure 3. *Focused ultrasound induces microbubble oscillation, microstreaming, and radiation forces that deform the endothelium and create shear stress. Inertial cavitation releases shock waves upon bubble collapse, collectively leading to localized stretching and disruption of endothelial tight junctions (arrowhead), enabling blood–brain barrier opening.⁹*

The temporary increase in BBB permeability facilitates localized delivery of therapeutics—including chemotherapy, antibodies, gene therapies, and cells—into brain

tissue.^{8-9,11,18} The BBB typically reseals within hours. MBs also lower the ultrasound intensity needed for effect, reducing the risk of skull and tissue damage.⁷ Real-time acoustic monitoring helps ensure safety by tracking cavitation activity during treatment.^{1,4-5,8}

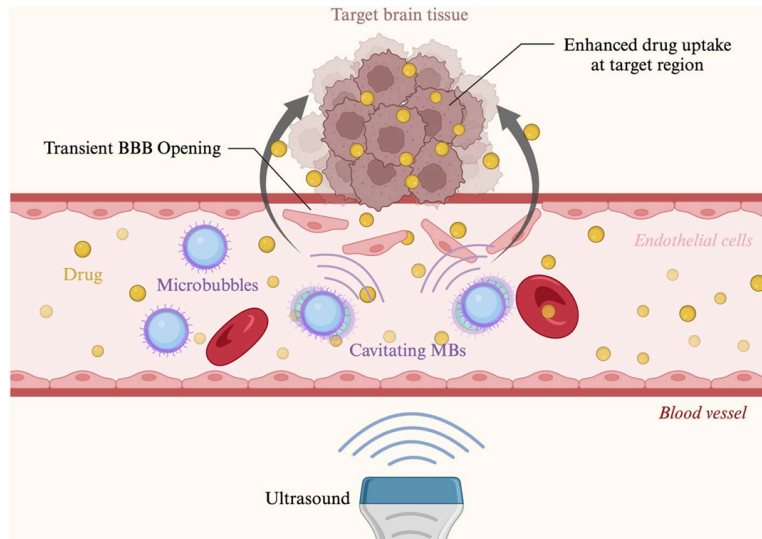


Figure 4. Schematic of FUS-induced BBB disruption. Focused ultrasound activates circulating microbubbles, generating shear stress that temporarily increases BBB permeability and enhances drug delivery to brain tissue.⁴

Comparative advantages

MRgFUS offers key advantages over both surgery and radiation therapy. As a non-invasive technique, it eliminates the need for craniotomy, thereby reducing risks of infection, bleeding, anesthesia, and prolonged recovery time.⁸ It also spares healthy brain tissue and leaves no scarring.⁹ Compared to radiation therapy, MRgFUS does not involve ionizing radiation, eliminating long-term risks like secondary tumors.⁸⁻⁹ It also supports repeatable treatments without cumulative toxicity, making it ideal for recurrent conditions or repeated blood-brain barrier (BBB) opening.⁸⁻⁹ As previously stated, real-time MRI guidance allows immediate feedback and precise thermal targeting, improving accuracy and minimizing collateral damage.^{8,15,17}

Applications

Brain tumors and brain cancers

Primary brain tumors

As explained earlier, thermal ablation uses HIFU to raise tumor tissue temperature, inducing coagulative necrosis while sparing surrounding brain tissue. MRI thermometry enables real-time temperature monitoring to ensure precision and safety. Coluccia et al. (2014) reported the first successful noninvasive thermal ablation of a brain tumor, specifically a recurrent glioblastoma, using MRgFUS in a 63-year-old patient, demonstrating the feasibility of this technique.¹⁶ Technological advancements now allow transcranial delivery without craniectomy (i.e., removing a portion of the skull without immediate replacement), and clinical trials are ongoing to assess safety and efficacy for recurrent gliomas.^{5,9}

In addition, low-intensity MRgFUS combined with microbubbles can transiently and locally disrupt the BBB allowing chemotherapeutic agents like temozolomide (TMZ) and liposomal doxorubicin (DOX) to reach tumor and peritumoral regions more effectively.¹ Mainprize et al. (2019) demonstrated this increased chemotherapy accumulation in targeted areas following BBB opening, confirmed by imaging and biochemical analysis.¹⁰

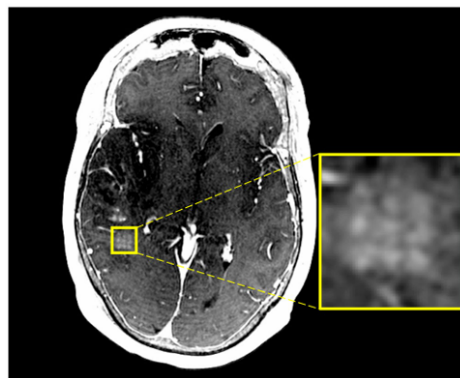


Figure 5. *Post-gadolinium axial T1-weighted MRI showing precise, grid-pattern contrast extravasation at the sonication sites with no signs of procedure-related edema.¹⁰*

Metastatic brain tumors

Brain metastases (i.e., cancer spread from a primary tumor to the brain) are a major cause of morbidity, with limited treatment options, especially for multiple or inoperable lesions.^{6,9} MRgFUS offers a noninvasive approach for palliative management by thermally ablating symptomatic metastases to reduce mass effect and relieve neurological symptoms, thereby improving quality of life.^{5,7-9,22} Additionally, MRgFUS-mediated BBB opening is being explored to enhance the delivery of systemic chemotherapies and immunotherapies, potentially improving local tumor control and survival.^{1-5,8-9,10,21,24,27} Ongoing clinical trials are evaluating its therapeutic potential in this setting.^{8,11}

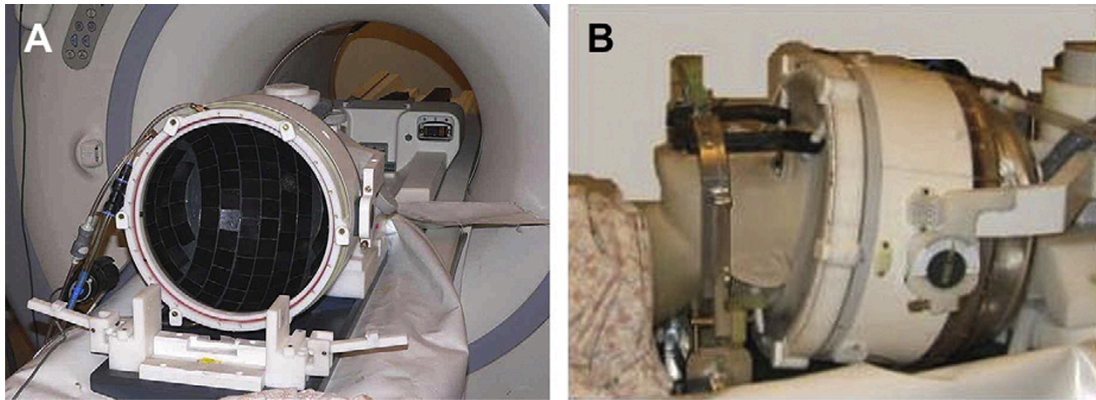


Figure 6. Exablate 4000 transcranial MRgFUS system. *A helmet-type hemispheric phased-array transducer (A) positioned over the patient's head with a degassed cold-water cap for coupling and cooling (B).*⁹

Other applications

Beyond brain tumors, MRgFUS has demonstrated efficacy and potential in treating a range of other neurological and non-neurological conditions.

Essential tremors

MRgFUS thalamotomy is a noninvasive, MRI-guided thermal ablation technique offering a safe and effective alternative to surgery for medication-refractory essential tremor.¹¹ Clinical trials

have demonstrated significant and lasting tremor reduction and improved quality of life, with submillimeter targeting precision comparable to or exceeding that of deep brain stimulation.²⁸ As previously stated, the FDA approved unilateral MRgFUS thalamotomy for essential tremor back in 2016, establishing it as a key clinical application.²⁵

Uterine fibroids

MRgFUS is an FDA-approved non-invasive treatment for symptomatic uterine fibroids.^{8,11} FUS energy is used to heat and ablate the fibroid tissue, leading to a reduction in symptoms such as pain and bleeding without the need for surgery.^{11,18} Once again, MRI guidance ensures precise targeting of the fibroids and monitoring of the thermal ablation process.²⁹

Breast tumors

MRgFUS is being investigated as a non-invasive treatment for localized, clinically non-palpable early-stage breast cancer, potentially offering an alternative to lumpectomy.^{7,19} Studies have shown that MRgFUS ablation can achieve significant tumor necrosis in breast cancer with good cosmetic results and preservation of tissue integrity.³⁰ Dedicated MRgFUS platforms for the breast have been developed, and clinical trials are underway to further evaluate the efficacy and safety of this approach.^{7,9}

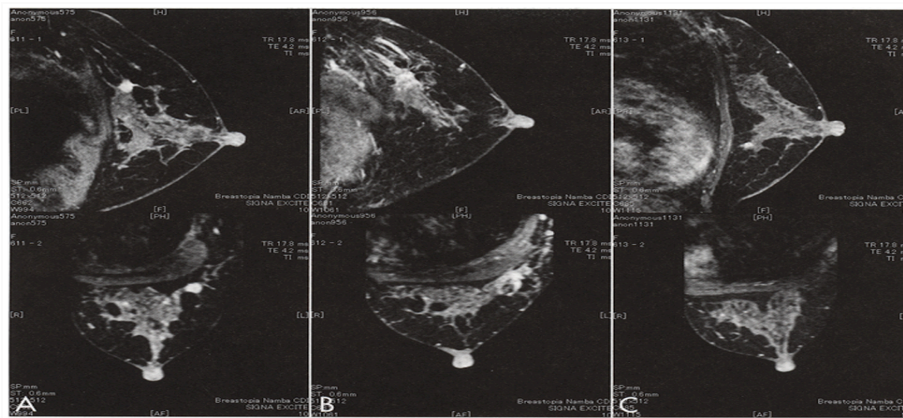


Figure 7. Progressive MRI changes in a non-recurrent case: (A) Initial 11 mm tumor in the upper-outer region. (B) One month post-MRgFUS. (C) One year post-MRgFUS.³⁰

Challenges

Technical limitations

As previously mentioned, transcranial MRgFUS is challenged by skull-induced attenuation, scattering, and phase aberrations due to variable bone thickness and density, which reduce the precision and efficacy of ultrasound delivery.²¹ While phased-array transducers and CT-based acoustic modeling help mitigate these effects, individualized corrections remain complex.¹¹ Skull absorption also leads to near-field heating, risking scalp burns and limiting energy delivery, especially in targets near the skull base.²¹ Cooling systems and larger transducer arrays help, but challenges persist for cortical applications.⁹

Clinical limitations

Large or infiltrative tumors are difficult to ablate completely, and lesions near critical structures or the skull present safety risks due to heating and beam distortion.^{9,11} Peripheral or bilateral brain targets remain less accessible¹¹. In BBB-opening applications, careful evaluation is required to avoid worsening peritumoral edema. Long-term outcomes and safety profiles are still under investigation.⁶

Economic challenges

The high cost of MRgFUS systems, requiring integrated MRI and specialized staff, restricts widespread adoption.¹⁸ Limited global availability and the need for contrast-enhanced MRI further limit access, especially for patients with renal impairment.² While early studies suggest cost benefits over invasive surgery for select indications, broader cost-effectiveness remains unproven.¹¹ Enhancing procedural efficiency, reducing equipment costs, and expanding access are key to increasing the clinical utility and sustainability of MRgFUS.^{2,11}

Future Directions

MRgFUS is expanding beyond its approved uses, showing promise in treating a wider range of neurological disorders and enabling novel non-thermal therapeutic strategies.

MRgFUS for other neurological diseases

Parkinson's disease

Parkinson's disease (PD) is a progressive neurodegenerative condition currently managed primarily with pharmacological therapies aimed at alleviating symptoms.⁸ However, these treatments often have limitations and may only temporarily delay disease progression.⁸ MRgFUS offers a non-invasive, targeted alternative for managing PD, particularly by enabling precise delivery of neurotrophic factors (e.g., GDNF) and gene therapies (e.g., AAV-GAD) to affected brain regions.⁸ Preclinical studies suggest benefits in preserving the nigrostriatal pathway (i.e., brain circuit responsible for controlling voluntary movements and body postural balance).³¹ Unlike invasive stereotactic methods, MRgFUS allows for repeated applications.⁸ Research also shows MRgFUS can modulate PD-related brain networks, with potential for broader symptom management beyond tremors.^{11,27}

Alzheimer's disease

Alzheimer's disease (AD) is another devastating neurodegenerative disorder characterized by the accumulation of amyloid-beta plaques and neurofibrillary tangles in the brain, leading to progressive cognitive decline.²⁷ Currently, there are no disease-modifying therapies widely available.^{8,27} MRgFUS enables transient, localized opening of the blood-brain barrier (BBB), improving delivery of therapeutic agents targeting amyloid-beta and tau pathology.^{9,27} Preclinical models demonstrate reduced plaque and tangle burden using antibody delivery. Notably, BBB

opening alone may facilitate clearance of pathological proteins.³ Ongoing trials are assessing MRgFUS in early AD patients for both therapeutic and diagnostic applications, including protein aggregate detection.^{1,8,27}

Non-thermal effects for neuromodulation and immunomodulation

Beyond thermal ablation and BBB opening for drug delivery, the non-thermal effects of MRgFUS are emerging as a significant area of future development, particularly in the realm of immunomodulation.^{8,11} At lower intensities and with pulsed-frequency application (pulsed FUS or pFUS), MRgFUS can induce reversible neuronal excitation and inhibition, a phenomenon known as neuromodulation.¹¹ This non-invasive modulation of neural activity holds potential for treating various neurological conditions, including epilepsy.¹¹ Research at institutions like the University of California Los Angeles (UCLA) is actively investigating the use of pulsed FUS for the treatment of epilepsy.¹¹ Furthermore, studies have shown that MRgFUS combined with microbubbles can influence the behavior of immune cells in the brain.³ In the context of GBM, a highly aggressive brain tumor, MRgFUS with microbubbles has been shown to promote the homing and differentiation of monocytes and induce the polarization of both monocytes and microglia towards a pro-inflammatory direction.³ This suggests that MRgFUS could potentially be employed as a stand-alone therapeutic strategy by stimulating an anti-tumor immune response or as an adjunct therapy to enhance the delivery and efficacy of chemotherapeutic agents for GBM.^{3,5} The ability of MRgFUS to facilitate the delivery of immunotherapies, such as antibodies, across the BBB for various neurological disorders, including Alzheimer's disease and potentially brain tumors, further underscores its potential in immunomodulation.^{3,11} The selective and reversible disruption of the BBB allows for focused delivery of immune-modulating agents directly to the brain parenchyma.²⁷

Conclusions

MRgFUS represents a promising breakthrough in the non-invasive treatment of brain tumors and cancers. This innovative technology combines the precision of MRI guidance with the therapeutic potential of focused ultrasound to address significant limitations in conventional treatment approaches. The dual mechanisms of action—thermal ablation via HIFU and mechanical disruption via LIFU—provide versatile therapeutic strategies for various tumor types, such as glioblastoma multiforme and metastatic lesions.

The historical progression of MRgFUS, from Lynn's initial demonstrations in the 1940s to modern clinical applications, illustrates the remarkable evolution of this technology. Through integration of MRI guidance, phased-array transducers, and skull aberration correction techniques, MRgFUS has transformed from an experimental concept to a clinically viable treatment modality offering distinct advantages over traditional surgical and radiotherapeutic interventions.

Despite these advancements, significant challenges persist. Technical limitations related to skull-induced heating and ultrasound refraction constrain treatment of peripheral tumors. Economic barriers, including high equipment costs and limited global availability, present additional hurdles. While early clinical trials demonstrate promising results, long-term efficacy and safety data across diverse patient populations remain insufficient.

Looking forward, the future of MRgFUS appears promising, with emerging applications extending beyond oncology to neurodegenerative disorders such as Parkinson's and Alzheimer's disease, while its potential for immunomodulation through non-thermal effects offers an exciting avenue for enhancing antitumor immune responses. As these challenges are systematically addressed, MRgFUS is projected to become an increasingly important component of the multimodal treatment against brain tumors and cancers.

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